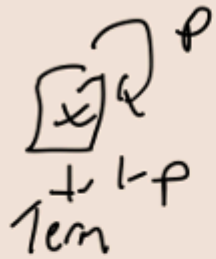
Q1.)

a.) The results would still be the same because the states can only be visited once.

1.)

b.)



+1 on all trans

x	+1	0
x	+2	x
x	+3	x
x	+4	x
x	+5	x
x	+6	x
x	+7	x
x	+8	x
x	+9	x
term	+10	x

first visit

10/1 = 10

every-visit

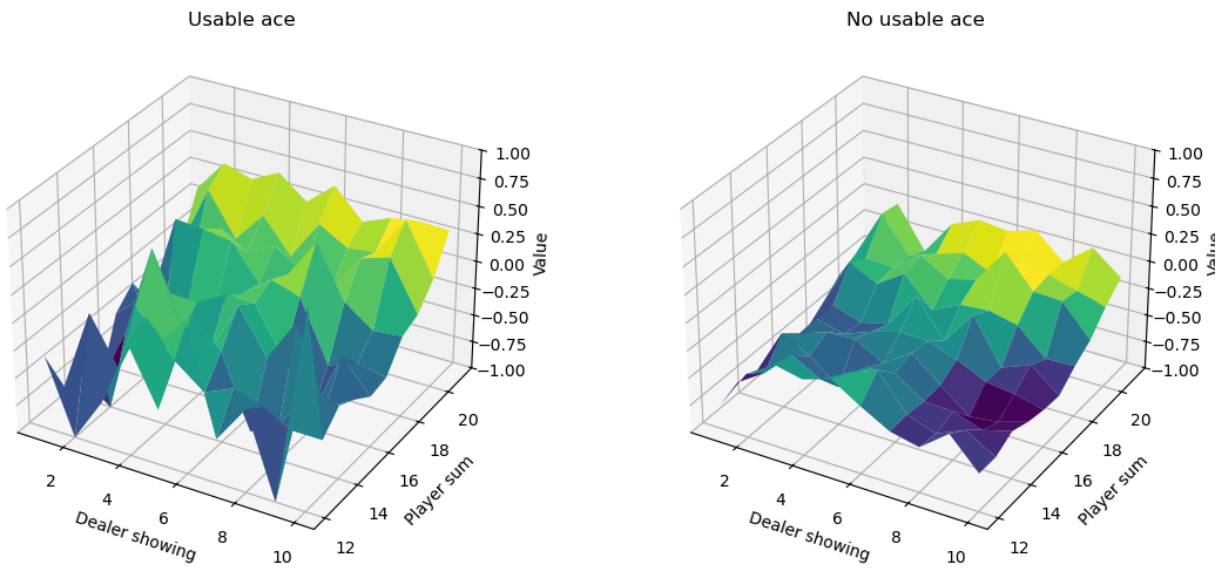
10+9+8+...+1/10 = 55

b.)

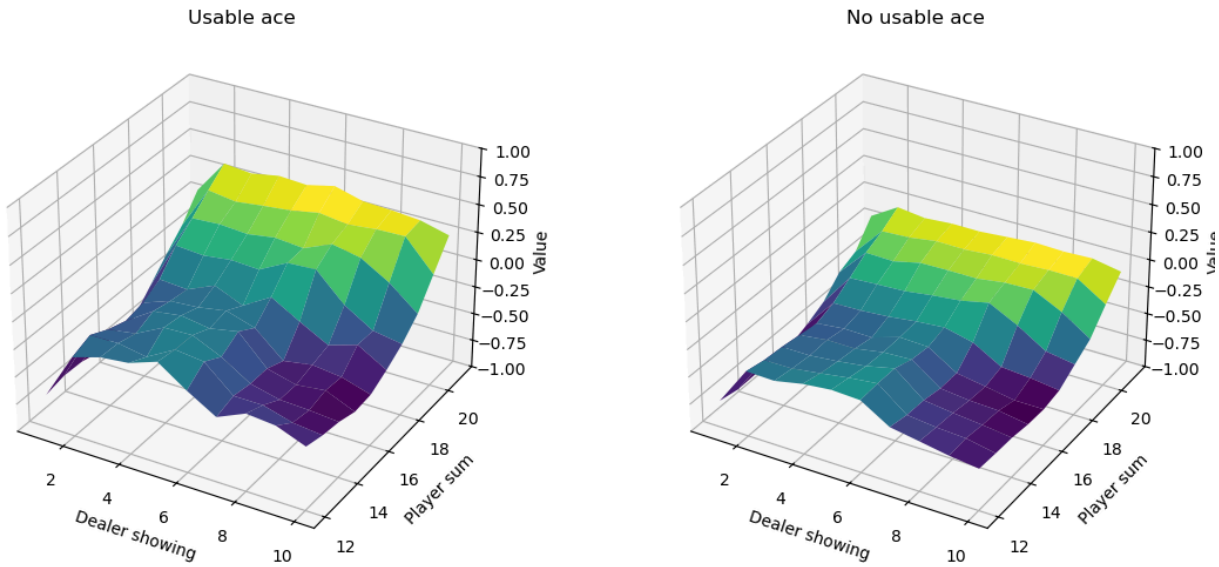
The period between the 5.5 got deleted. It should say 5.5

Q2.)

a.)

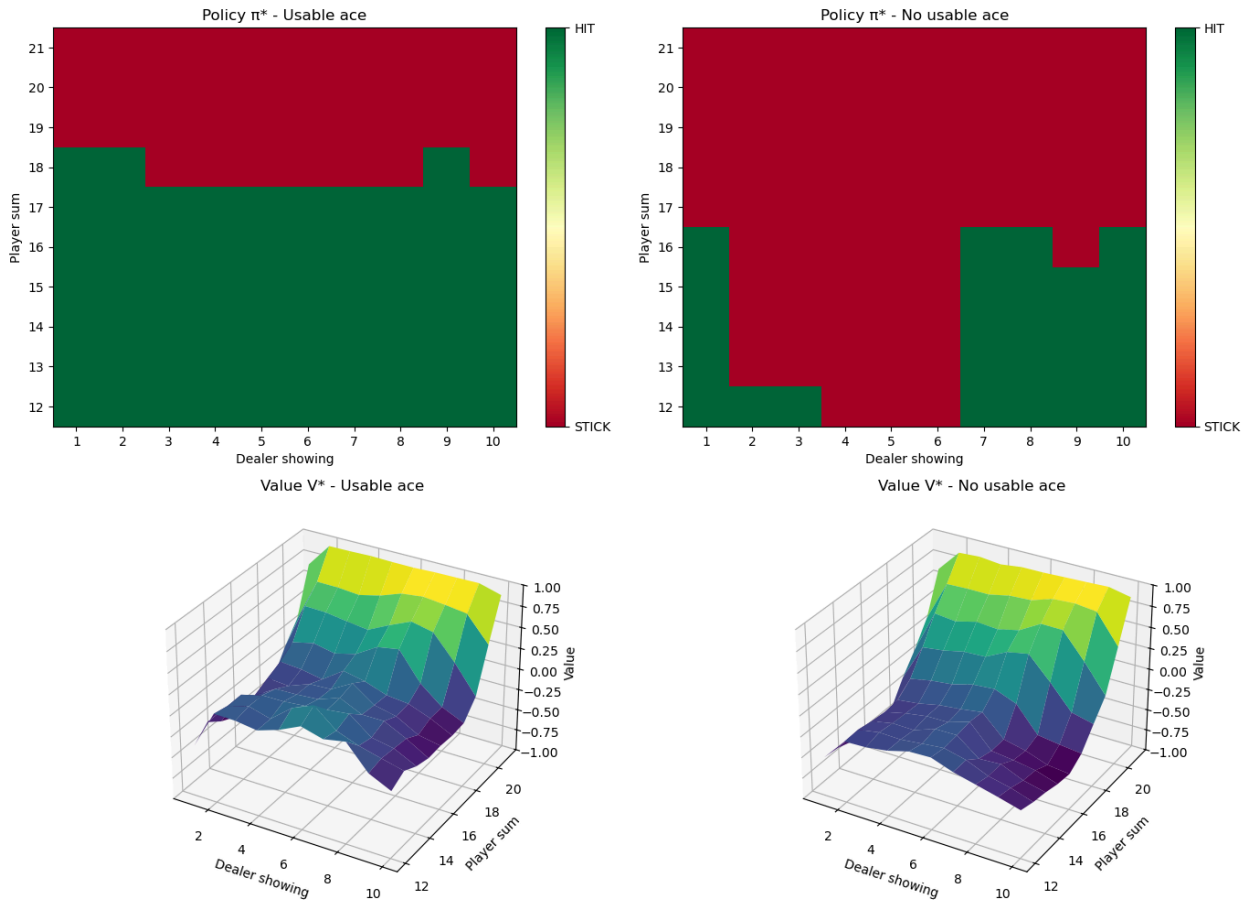


ten thousand



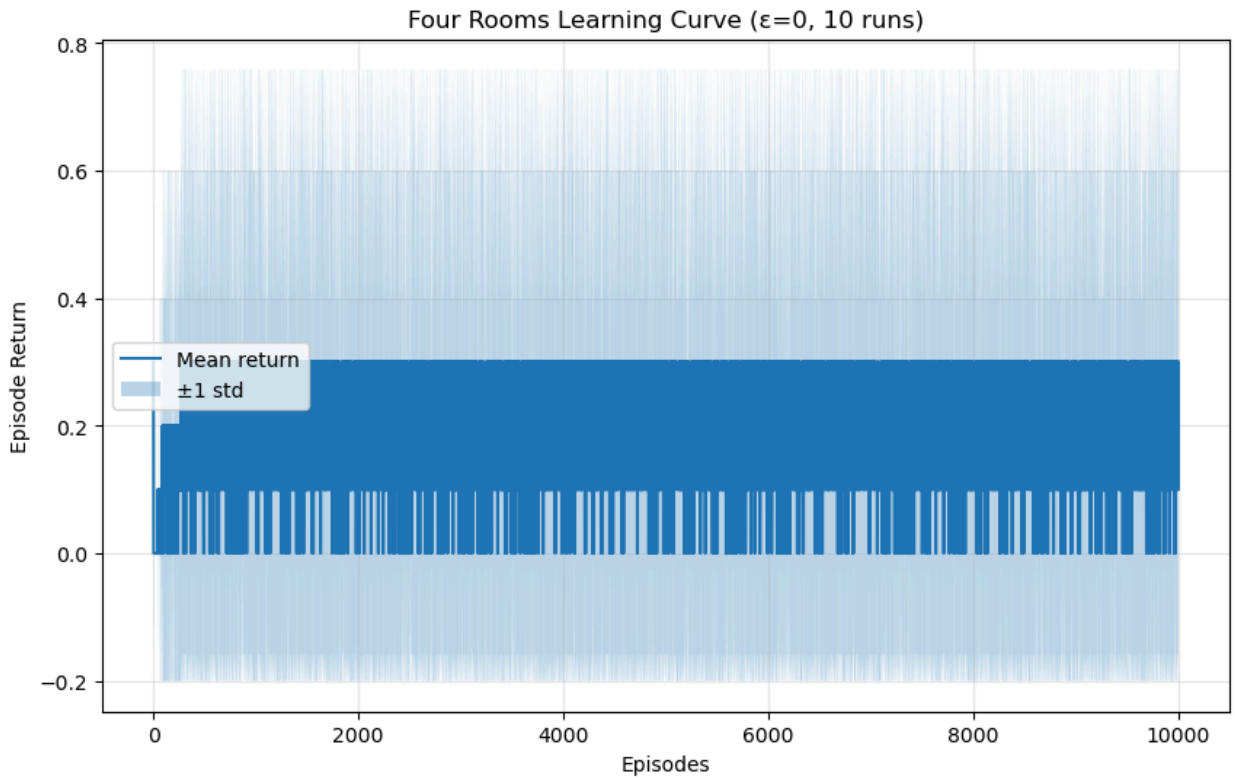
Half a million

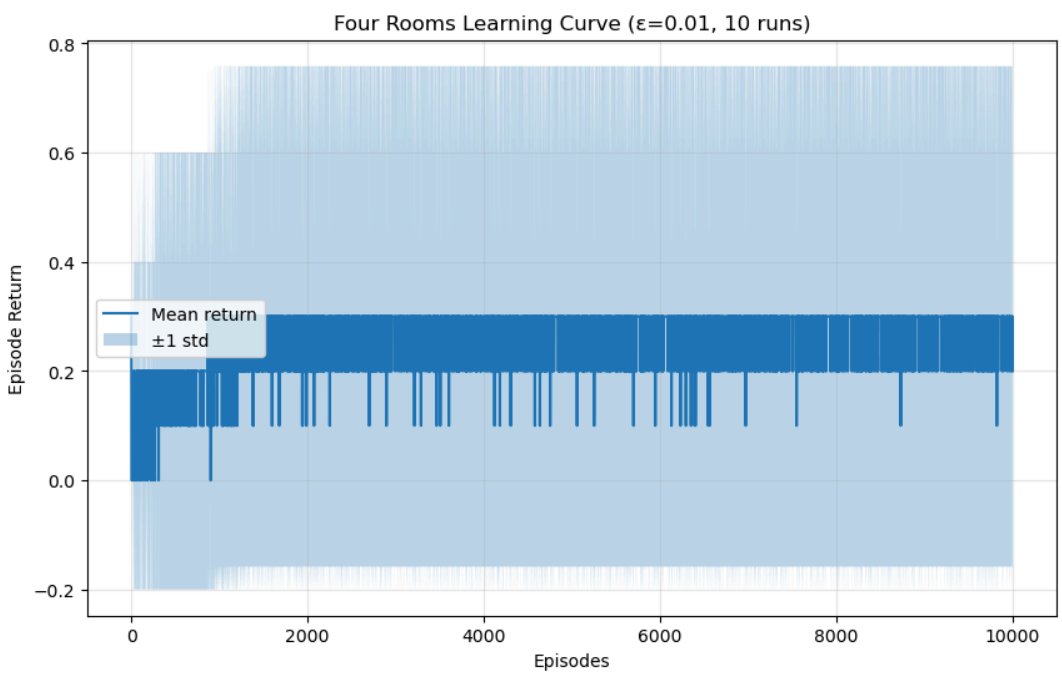
b.)

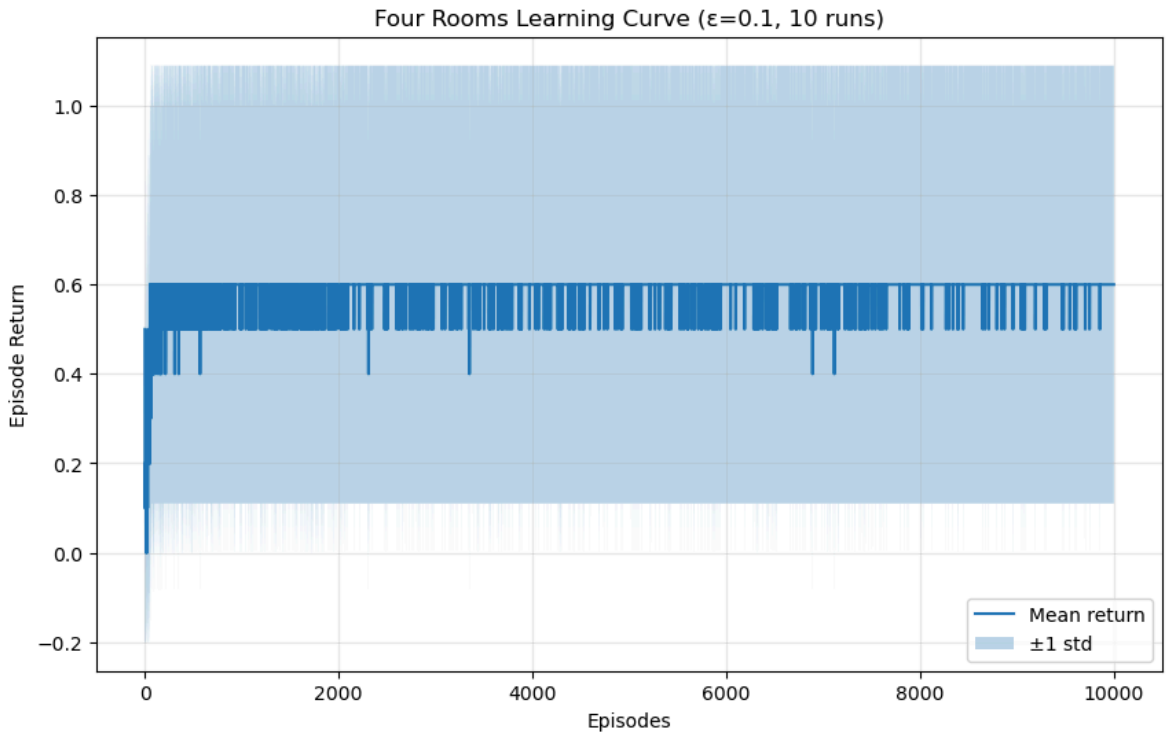

Policy π* - Usable ace
Policy π* - No usable ace
Player sum
Dealer showing
HIT
STICK
Value V* - Usable ace
Value V* - No usable ace
Value
Player sum
Dealer showing

Q3.)

a.) plot:


Four Rooms Learning Curve (ε=0, 10 runs)
Episode Return
Episodes
Mean return
±1 std
0.8
0.6
0.4
0.2
0.0
−0.2
0
2000
4000
6000
8000
10000


Four Rooms Learning Curve (ε=0.01, 10 runs)
Episode Return
Episodes
Mean return
±1 std
0.8
0.6
0.4
0.2
0.0
−0.2
0
2000
4000
6000
8000
10000



b.) By not doing an exploring start my monte carlo policy barely explores if I could even call it exploring. I get little to no episodic returns while I use ε = 0 and I stay in that return range. However if I set it to 10% my returns shoot up immediately and I stay in this higher return range for the rest of my episodes.

Q4.)

a.)

4.)

a) $$Q_n = \frac{\sum_{k=1}^{n-1} W_k G_k}{\sum_{k=1}^{n-1} W_k} \qquad C_n = \sum_{k=1}^{n-1} W_k$$

$$Q_n = \frac{1}{C_n} \sum_{k=1}^{n-1} W_k G_k$$

$$Q_{n+1} = \frac{\sum_{k=1}^{n-1} W_k G_k}{\sum_{k=1}^{n-1} W_k} = \frac{W_n G_n + \sum_{k=1}^{n-1} W_k G_k}{C_n + W_n}$$

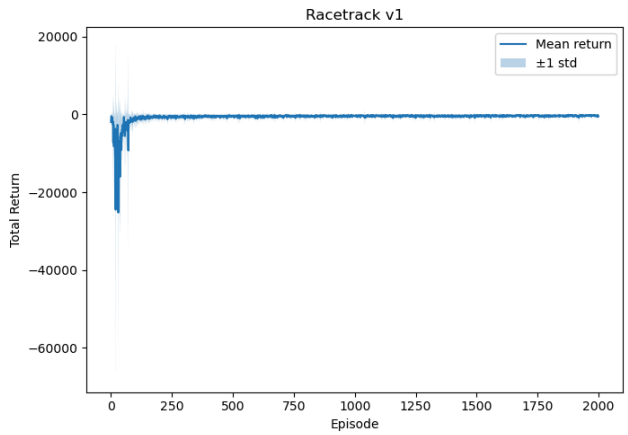
$$Q_{n+1} = \frac{W_n G_n + C_n Q_n}{C_n + W_n} = \frac{(C_n + W_n) Q_n - W_n Q_n + W_n G_n}{C_n + W_n}$$

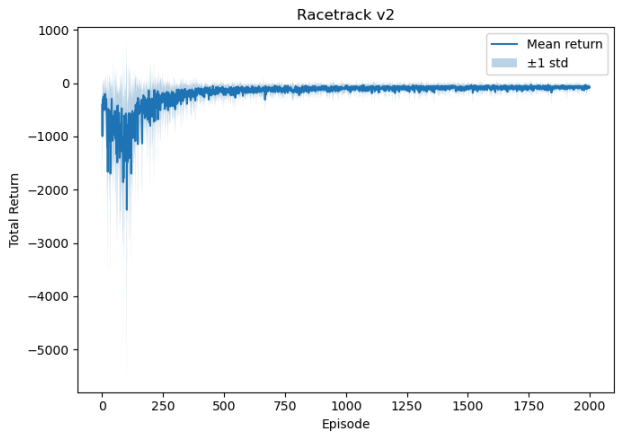
$$Q_{n+1} = Q_n + \frac{W_n}{C_{n+1}} (G_n - Q_n) \qquad C_{n+1} = C_n + W_n$$

b.) Since π is greedy, π(At|St) is always either 0 or 1. The loop already bails out when it's 0, so by the time you're updating W, you're guaranteed π(At|St) = 1. That means the full ratio π(At|St)/b(At|St) just becomes 1/b(At|St), so that's all you need to multiply by.

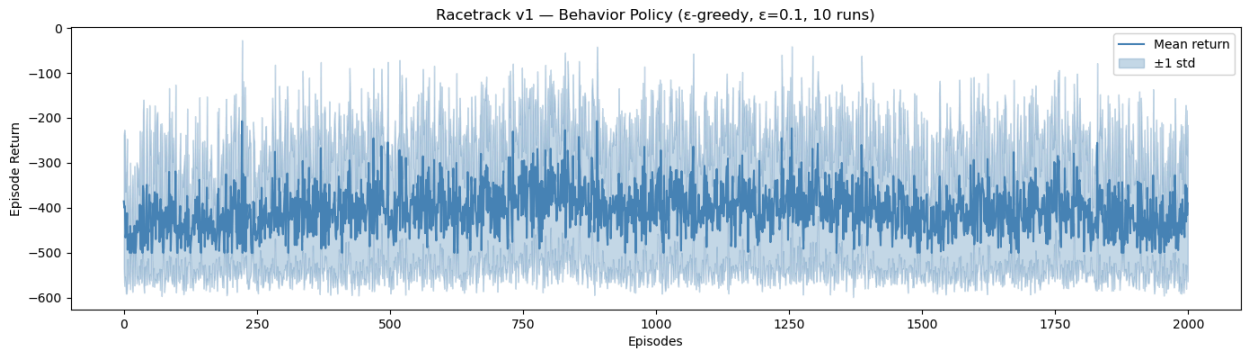
Q5.)

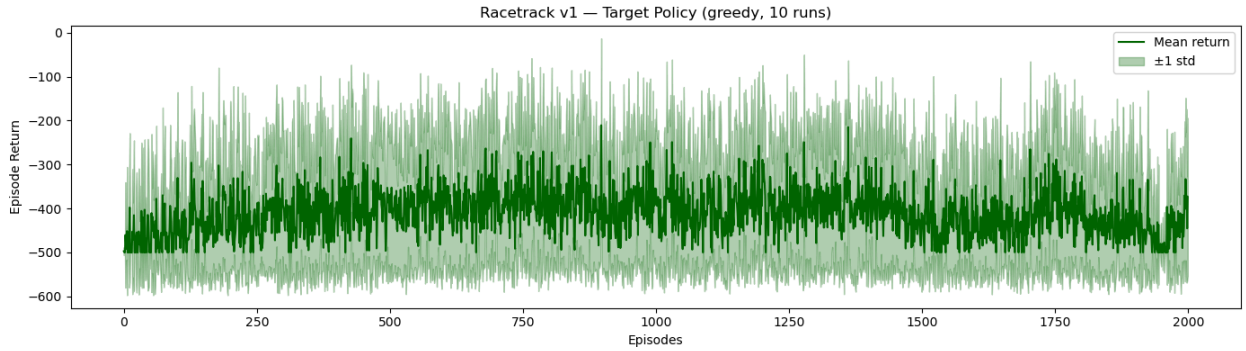
a.)

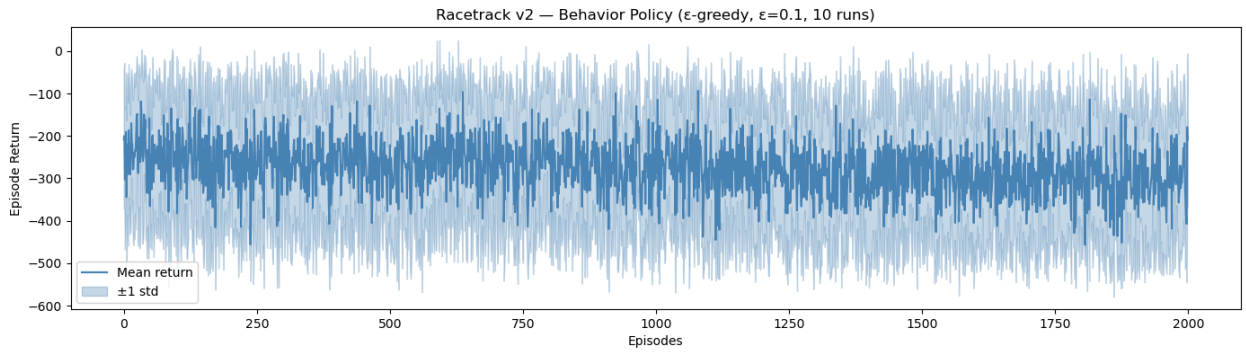

Racetrack v1
Mean return
±1 std
Total Return
Episode

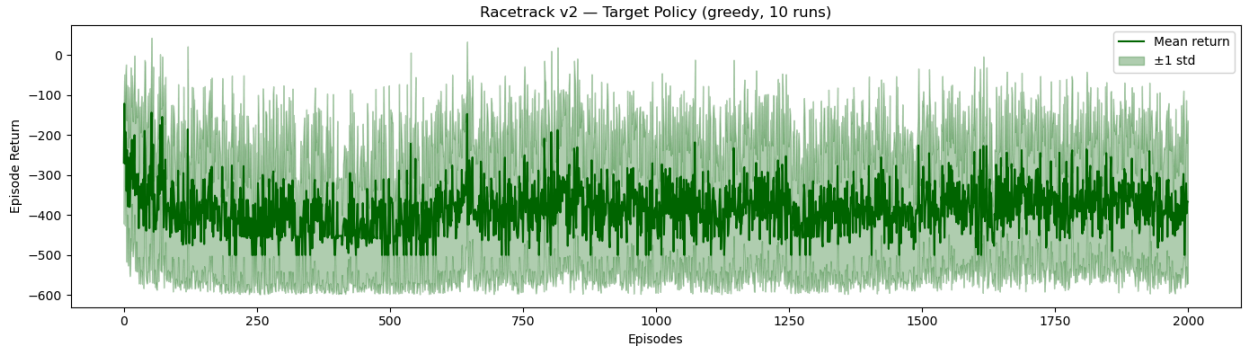

Racetrack v2
Mean return
±1 std
Total Return
Episode

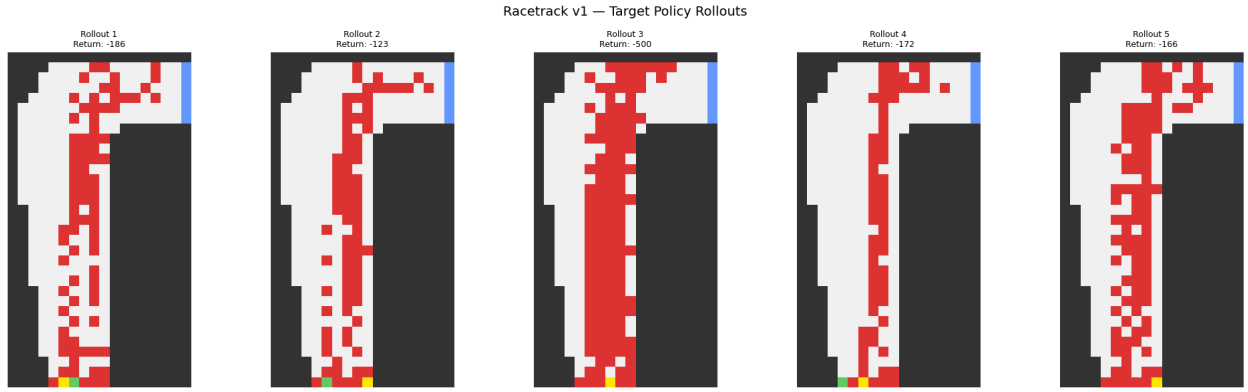
b.)

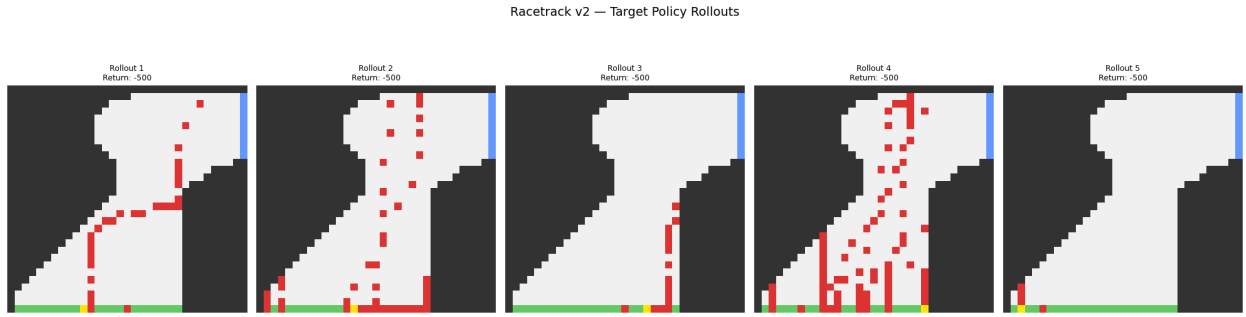

Racetrack v1 — Behavior Policy (ε-greedy, ε=0.1, 10 runs)
Mean return
±1 std
Episode Return
Episodes


Racetrack v1 — Target Policy (greedy, 10 runs)
Mean return
±1 std
Episode Return
Episodes









c.) The main thing I observed is that off-policy for racetrack takes incredibly long to step through. Off-policy hardly ever seems to reach the goal and if not truncated in steps will take so long it is not worth running. It is better to use on-policy for racetrack.